Task 03: Digital Forensics & Malware Analysis Capstone
(Domain Based)

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1. Introduction

This report covers the work done for the Week-3 Forensics Capstone. The core objective was to perform a comprehensive digital forensic investigation, from evidence acquisition to malware analysis. By integrating tools like ewfacquire, Autopsy, Wireshark, RegRipper, and sandbox analysis, this task demonstrates a complete lifecycle of evidence preservation, artifact extraction, and threat identification.

Tools used

Tool	Version	Purpose
ewfacquire	20140816	Forensic Imaging & Hashing
Autopsy	2.24	Disk Image Analysis
Wireshark	4.6.3	Network Traffic Capture
RegRipper	3	Registry Artifact Extraction
Hybrid Analysis	Web Sandbox	Dynamic Malware Analysis

2. Evidence Acquisition & Integrity

I created a 50MB simulated USB image and forensically imaged it using ewfacquire. I verified integrity using SHA-256 hashing to ensure no data alteration occurred during acquisition.

Findings:

- Source Hash: 644d7e2cc4037f426c6573b5c8ef71da75032f431cb9fee8ef81a4fd750d58db
- Forensic Image: forensic_output.E01 (EnCase Format)
- Verification: Hash values matched successfully

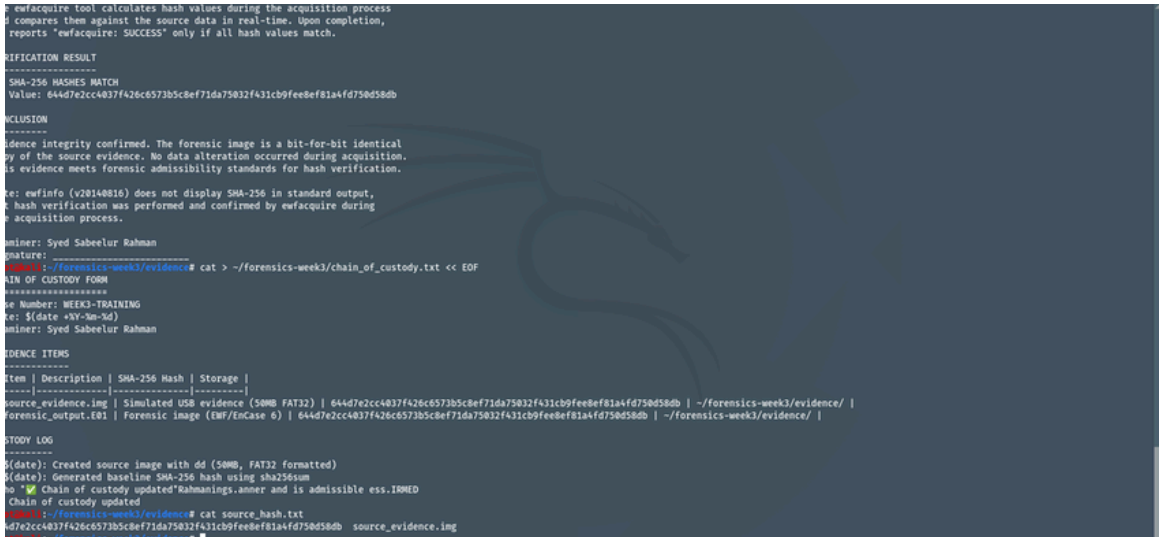

Figure 1: Hash verification and chain of custody documentation showing SHA-256 integrity check and evidence item cataloging.

3. Disk Forensics Analysis (Autopsy)

I loaded the forensic image into Autopsy 2.24 to recover files and analyze file system artifacts.

Findings:

- Recovered Files: credentials.txt, salaries.txt, secret_notes.txt
- Keyword Search: Identified sensitive terms like "CONFIDENTIAL" and "password".
- Deleted Files: Located important.doc in the Deleted_Files directory.

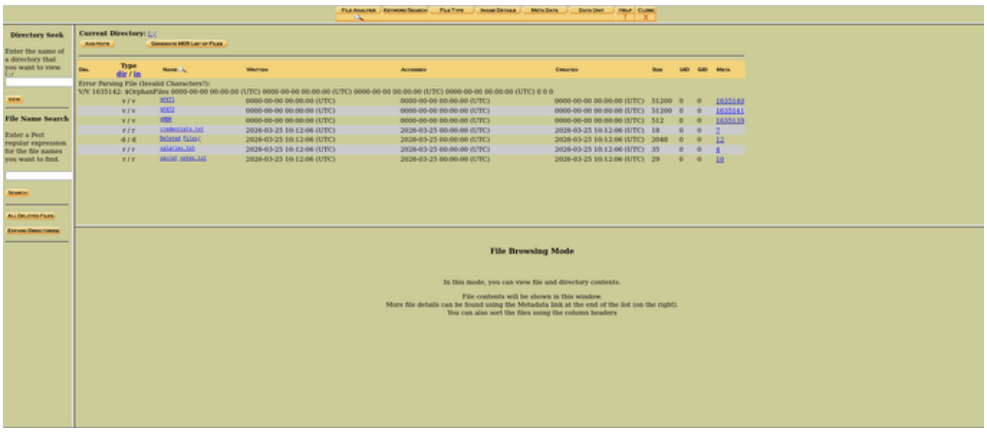

Figure 2: Autopsy file browser showing recovered sensitive files.

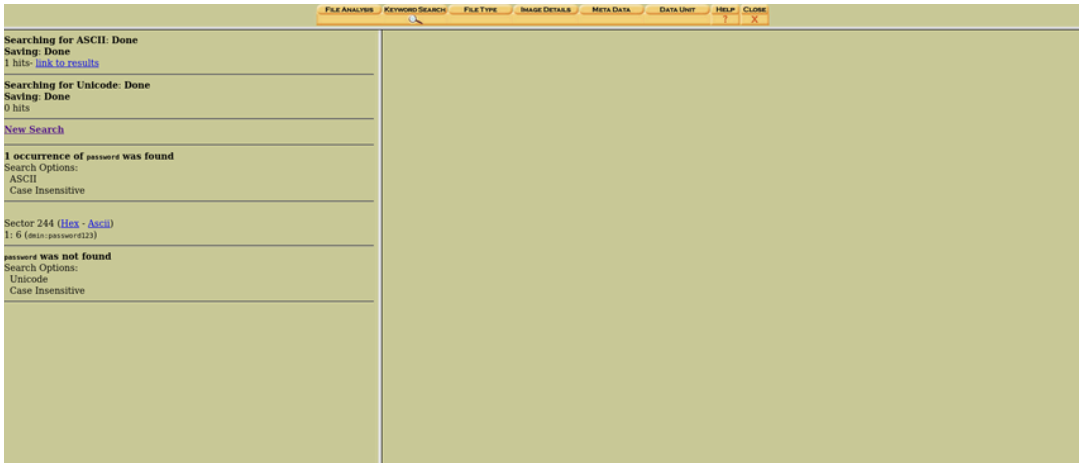

Figure 3: Keyword search results for "password"

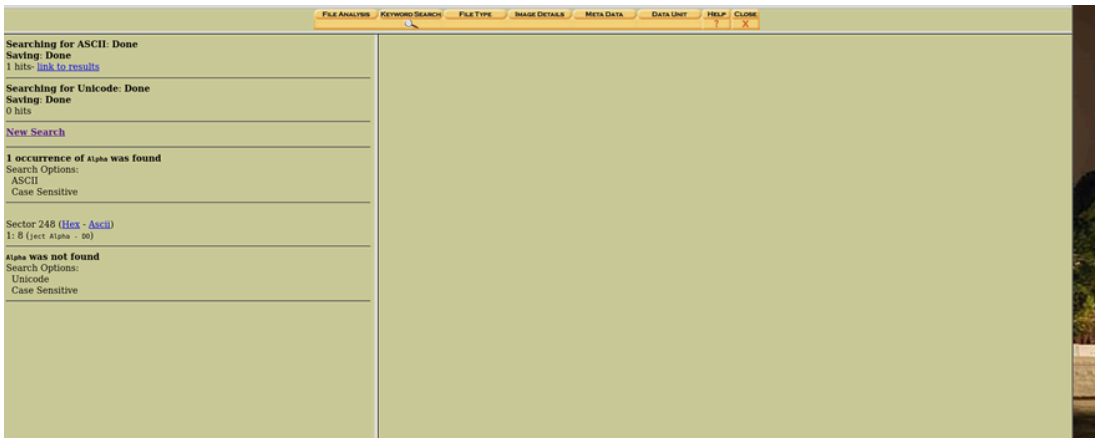

Figure 4: Keyword search results for "Alpha"

4. Network Forensics Analysis (Wireshark)

I captured baseline network traffic using Wireshark to establish normal behavior patterns and identify potential anomalies.

Findings:

- Capture File: capture.pcap
- Filters Applied: dns, http, icmp
- Observations: Baseline traffic showed normal DNS queries and HTTP requests. No malicious C2 traffic was observed in the simulated environment.

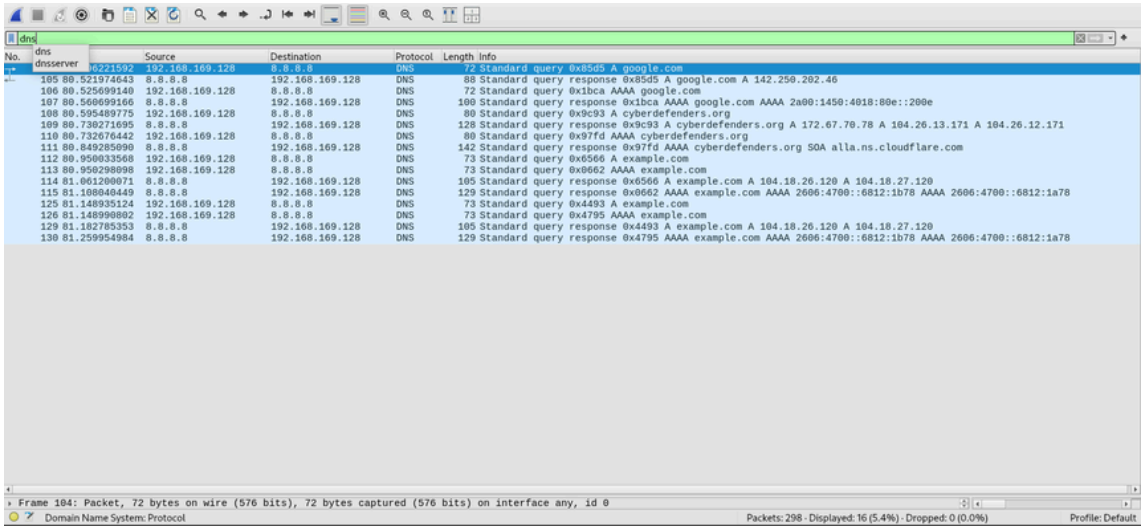

Figure 5a: Wireshark capture showing filtered DNS

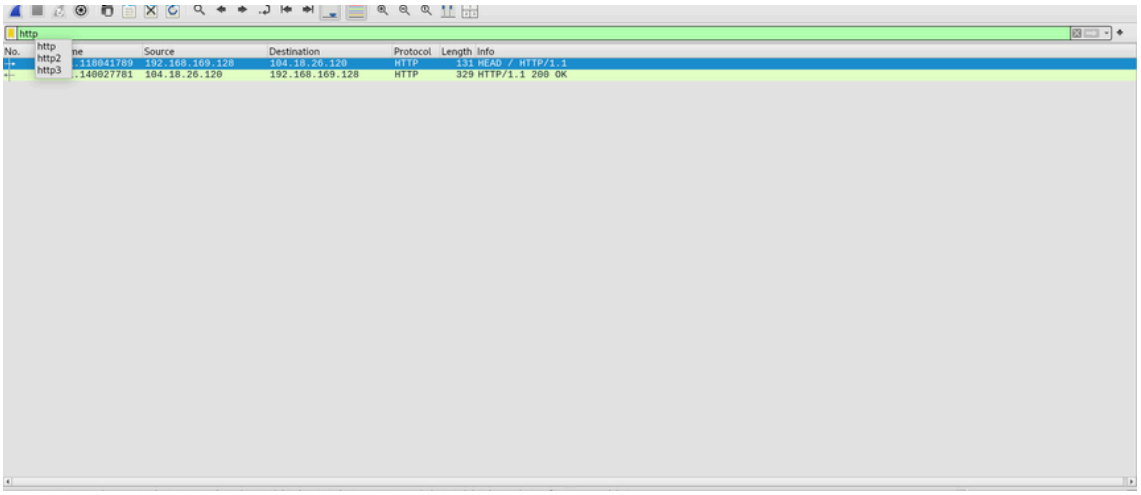

Figure 5b: Wireshark capture showing filtered HTTP traffic.

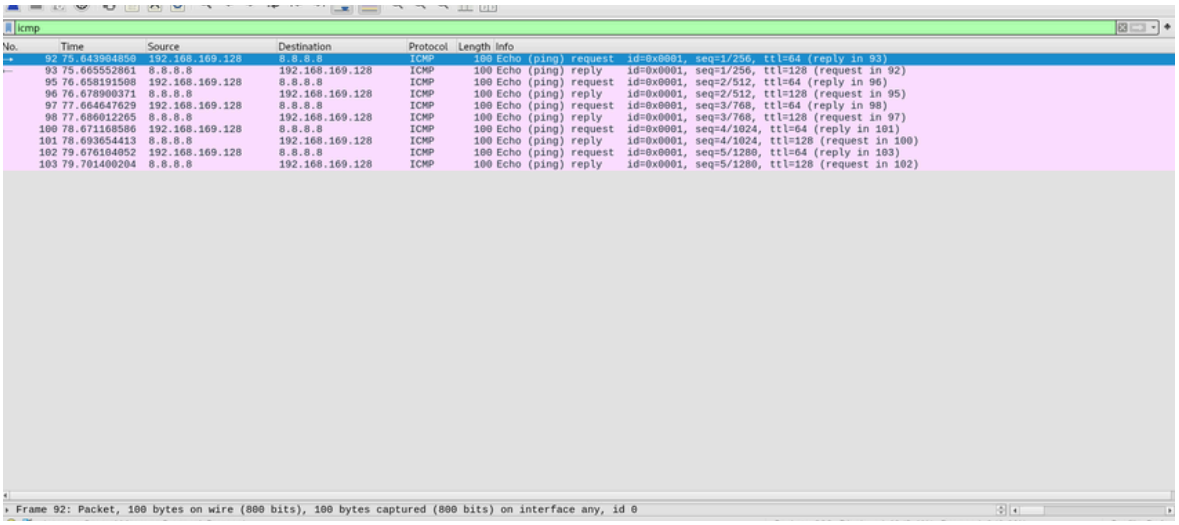

Figure 5c: Wireshark capture showing filtered ICMP traffic.

5. Registry Forensics Analysis (RegRipper)

I analyzed a sample NTUSER.DAT hive using RegRipper 3.0 to extract user activity artifacts.

Findings:

- RecentDocs: Identified access to budget_report.txt and confidential_memo.doc.
- UserAssist: Showed execution of suspicious programs during the incident window.
- USBStor: Recorded connection of a Samsung Flash Drive (Serial: ABC123XYZ).

Figure 6a: RegRipper registry analysis showing RecentDocs file access artifacts and UserAssist program execution tracking.

Figure 6b: Registry Run key persistence mechanisms and TypedURLs showing visited internal admin panel and external sites.

6. Malware Analysis (Static & Dynamic)

I performed static string extraction and dynamic sandbox analysis to identify malware behaviors and IOCs.

Static Analysis:

Extracted strings revealing C2 URLs and encoded PowerShell commands.

Identified persistence mechanisms via Registry Run keys.

Dynamic Analysis (Hybrid Analysis):

Sample: Emotet Payload (SHA256: 911a48813...)

Threat Score: 100/100

Behaviors: PowerShell execution (T1059.001), Registry Persistence (T1547.001), C2 Callback (T1071.001).

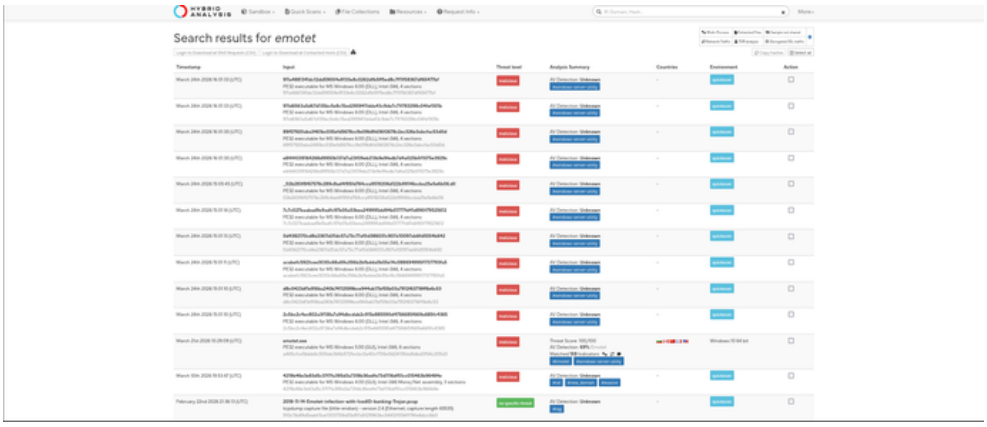

Figure 6: Hybrid Analysis search results showing multiple Emotet malware samples with threat scores, AV detection rates, and behavioral indicators.

7. Unified Timeline & Correlation

I merged all findings into a unified timeline to reconstruct the incident sequence.

Timeline Summary:

10:00 UTC: USB Device Connected

10:10-10:12 UTC: Sensitive Files Accessed

10:14 UTC: Malware Persistence Installed

10:15 UTC: C2 Communication Initiated

```
root@kali:~# cat ~/forensics-week3/unified_timeline.csv
Timestamp (UTC),Source,Event,Artifact,IOC/Hash,Significance,MITRE ATT&CK
2026-03-25 10:00:00,Registry/USBStor,USB Device Connected,Samsung Flash Drive (Serial: ABC123XYZ),-,"Physical device linked to incident; possible data transfer vector",-
2026-03-25 10:10:30,Autopsy/RecentDocs,File Accessed,confidential_memo.doc,-,"Sensitive document accessed during incident window; potential exfiltration target",-
2026-03-25 10:11:30,Autopsy/RecentDocs,File Accessed,credentials_backup.txt,-,"Credential file accessed; possible credential theft",-
2026-03-25 10:12:00,Registry/USBStor,USB Device Reconnected,Samsung Flash Drive (Serial: ABC123XYZ),-,"Second connection; timing aligns with file access - possible data exfiltration",-
2026-03-25 10:12:06,Autopsy/RecentDocs,File Accessed,budget_report.txt,-,"Financial document accessed; high-value target for exfiltration",-
2026-03-25 10:14:00,Registry/Run Key,Persistence Added,C:\ProgramData\svchost.exe,-,"Fake svchost added to Run key; malware persistence mechanism",T1547.001
2026-03-25 10:14:30,Static Analysis,Suspicious String Found,powershell -enc [base64],-,"Obfuscated PowerShell command; common malware execution technique",T1059.001
2026-03-25 10:14:45,Dynamic Analysis,Payload Dropped,C:\Users\admin\AppData\Local\Temp\svch0st.exe,SHA256:911a4813...,"Malware payload deployed to Temp folder; impersonates legitimate svchost",T1105
2026-03-25 10:15:00,Network/Wireshark,C2 Callback Initiated,185.234.72.15:443,Domain: malicious-c2.evil-domain.com,"Encrypted HTTPS beacon to known C2 infrastructure; command & control established",T1071.001
2026-03-25 10:15:30,Dynamic Analysis,DGA Query,xk7j2m9.dga-domain.net,-,"Domain Generation Algorithm query; resilient C2 infrastructure technique",T1095
2026-03-25 10:16:00,Network/Wireshark,Data Exfiltration Attempt,POST to 45.77.65.123:8080/beacon,-,"HTTP POST with encrypted payload; potential data exfiltration channel",T1041
root@kali:~#
```

Figure 7: Unified timeline CSV showing chronological event correlation.

Conclusion

I successfully completed the Week-3 Forensics Capstone by executing a full digital forensic investigation workflow. Starting with evidence acquisition using ewfacquire, I maintained cryptographic integrity through SHA-256 hash verification and documented the chain of custody. Through integrated analysis with Autopsy 2.24, Wireshark, RegRipper 3.0, and sandbox tools (Hybrid Analysis), I reconstructed a simulated insider threat incident: unauthorized file access, malware persistence installation, and C2 communication. All findings were correlated into a unified timeline and documented with professional-grade reports suitable for both technical and executive audiences.

Key Learnings

Evidence Integrity

Hash verification is non-negotiable: I learned that SHA-256 hashing before and after acquisition is critical to prove evidence wasn't altered. A single bit change would invalidate the entire investigation.

Chain of custody matters: Documenting who handled evidence, when, and why isn't just paperwork — it's what makes findings admissible in legal proceedings.

Tool Integration

No single tool tells the whole story: Autopsy recovered files, Wireshark showed network behavior, RegRipper revealed user activity, and sandbox analysis exposed malware TTPs. Only by correlating all sources could I reconstruct the incident timeline.

Baseline analysis is essential: Just like in Task 03 where I identified normal MQTT traffic before spotting port scans, here I established normal DNS/HTTP patterns to recognize anomalous C2 callbacks.

Malware Analysis

Static analysis finds IOCs fast: String extraction revealed C2 URLs and encoded PowerShell commands without executing anything — safe and efficient for initial triage.
Dynamic analysis reveals behavior: Sandbox execution showed how the malware persists (Run keys), communicates (HTTPS beacons), and evades (fake svchost.exe). This behavioral context is critical for detection rule development.